



AWS Secrets Manager

Lecture Slides

These lecture slides are provided for personal and non-commercial use only

Please do not redistribute or upload these lecture slides elsewhere.

Good luck on your exam!

Secrets Manager



Protect secrets needed to access your applications and services.
Easily rotate, manage, and retrieve database credentials, API keys,
and other secrets throughout their lifecycle

Introduction to Secrets Manager



Secrets Manager is mainly used to store and automatically rotate database credentials.



RDS



Redshift



DocumentDB



Other Databases



Key/Value

Select secret type [Info](#)

☐ Credentials for RDS database	☐ Credentials for Redshift cluster	☐ Credentials for DocumentDB database
☒ Credentials for other database	☐ Other type of secrets (e.g. API key)	

Select which database this secret will access [Info](#)

☐ MariaDB	☐ MySQL	☒ PostgreSQL	☐ ORACLE DATABASE	☐ Microsoft SQL Server
-------------------------------	-----------------------------	---	---------------------------------------	--

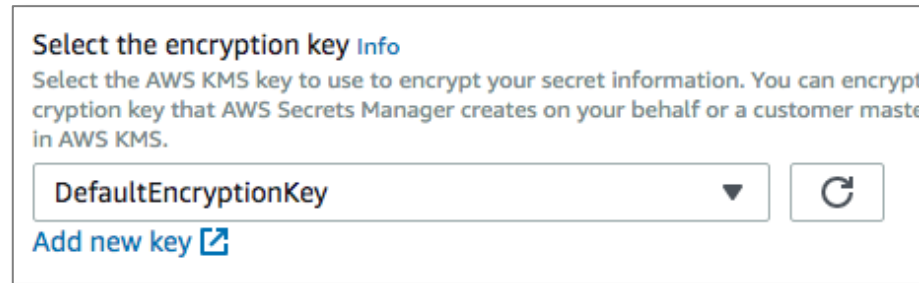
Server address	Database name	Port
127.0.0.1	praxis	5432

Specify the key/value pairs to be stored in this secret [Info](#)

Secret key/value	Plaintext
self-destruct	000destruct0
+ Add row	

Introduction to Secrets Manager

Enforces encryption
at rest by using KMS



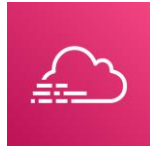
Select the encryption key [Info](#)
Select the AWS KMS key to use to encrypt your secret information. You can encrypt with a customer master key (CMK) that AWS Secrets Manager creates on your behalf or a customer master key (CMK) that you create in AWS KMS.

DefaultEncryptionKey ▼ 

[Add new key](#) 

Pricing

- \$0.50 per secret per month
- \$0.05 per 10,000 API calls



CloudTrail can monitor credentials access in case you need to audit.

Secrets Manager – Automatic rotation

You can setup automatic rotation for any database credentials.

You can rotate up to 365 days (1 year)

Rotation is performed via a Lambda function

You can rotate the password for the superuser or for a developer programmatically accessing the database.

The screenshot shows the 'Configure automatic rotation' page in the AWS console. It includes options to enable or disable rotation, a dropdown for the rotation interval (30 days), and radio buttons for creating a new Lambda function or using an existing one. A text input field shows the function name 'exampro-key-rotation'. At the bottom, there are radio buttons to select which secret to use for rotation. Two callout boxes are present: one showing the interval dropdown menu with options 30, 60, 90 days, and Custom; the other showing the custom interval input set to 365 days. Red arrows point from the explanatory text on the left to the corresponding UI elements.

Configure automatic rotation - optional [Info](#)
Configure AWS Secrets Manager to rotate this secret automatically. Read the [getting started guide](#) on rotation.

☐ Disable automatic rotation
Recommended when your applications are using this secret and have not been updated to use AWS Secrets Manager.

☒ Enable automatic rotation
Recommended when your applications are not using this secret yet.

Select rotation interval [Info](#)
This secret will be rotated based on the schedule you determine.

30 days ▼
Maximum 365 days

☒ Create a new Lambda function to perform rotation

☐ Use an existing Lambda function to perform rotation

New AWS Lambda function name
Give your Lambda function a name. The prefix SecretsManager will be added to the name.

SecretsManager exampro-key-rotation

Function name must contain only alphanumeric characters, hyphens, or underscores. Do not exceed 64 characters.

Select which secret will be used to perform the rotation [Info](#)

☒ Use this secret
Use this option if you are storing a superuser.

☐ Use a secret that I have previously stored in AWS Secrets Manager
Use this option if you are storing a user who will access the database programmatically. Secrets Manager will use a previously stored superuser to execute rotation.

Select rotation interval [Info](#)
This secret will be rotated based on the schedule you determine.

30 days ▼
30 days
60 days
90 days
Custom

Select rotation interval [Info](#)
This secret will be rotated based on the schedule you determine.

Custom ▼ 365 days
Maximum 365 days

Secrets Manager – CLI

```
aws secretsmanager describe-secret --secret-id enterprise/ShipDatabase
```

```
{
  "ARN": "arn:aws:secretsmanager:region:123456789012:secret:enterprise/ShipDatabase-ji0b0V",
  "Name": "enterprise/ShipDatabase",
  "Description": "Core Database",
  "LastChangedDate": 1522680794.8,
  "LastAccessedDate": 1522627200.0,
  "VersionIdsToStages": {
    "EXAMPLE1-90ab-cdef-fedc-ba987EXAMPLE": [
      "AWSCURRENT"
    ]
  }
}
```

Secrets Manager – CLI

```
aws secretsmanager get-secret-value --secret-id enterprise/ShipDatabase --version-stage AWSCURRENT
```

```
{  
  "ARN": "arn:aws:secretsmanager:region:123456789012:secret:enterprise/ShipDatabase-ji0b0V",  
  "Name": "enterprise/ShipDatabase",  
  "VersionId": "EXAMPLE1-90ab-cdef-fedc-ba987EXAMPLE",  
  "SecretString": "{\"username\":\"worf\",\"password\":\"delta-omega-beta\"}",  
  "VersionStages": [  
    "AWSCURRENT"  
  ],  
  "CreateDate": 1522680764.668  
}
```

Secrets Manager – Use Case

